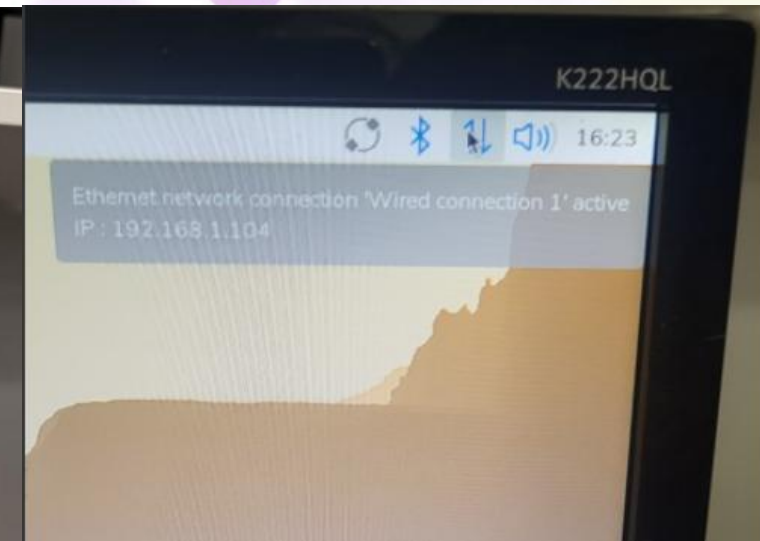
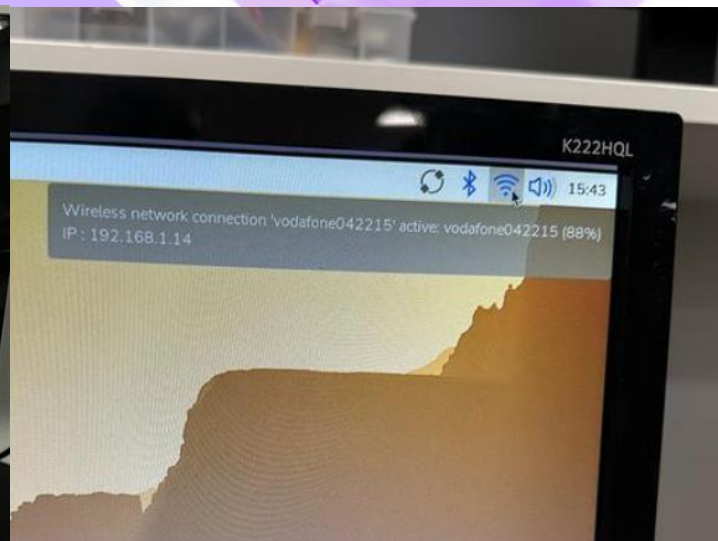
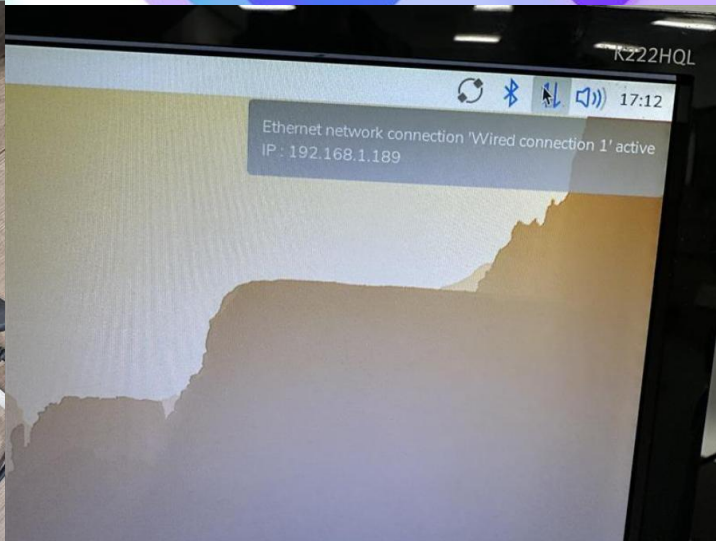


Unit 11 Assignment 2 Task 2

OLEKSANDR HANYSH

CONNECT (Router + IP from DHCP)



THREE USERS

PC1 - 1 USER

PC2 - 2 USER

PC3 - 3 USER

```
user1@raspberrypi1: ~/ShareRoot
File Edit Tabs Help
user1@raspberrypi1:~$ whoami
user1
user1@raspberrypi1:~$ hostname
raspberrypi1
user1@raspberrypi1:~$ mkdir ~/ShareRoot
user1@raspberrypi1:~$ cd ~/ShareRoot/
user1@raspberrypi1:~/ShareRoot$ mkdir Folder1 Folder2 Folder3 Folder4 Folder5
user1@raspberrypi1:~/ShareRoot$ touch Folder1/doc1.txt Folder1/doc2.txt Folder2/doc3.txt Folder3/doc4.txt Folder4/doc5.txt Folder5/doc6.txt Folder5/doc7.txt
```

```
user5@raspberrypi5: ~
user5@raspberrypi5: ~
user5@raspberrypi5: ~
File Edit Tabs Help
user5@raspberrypi5:~$ whoami
user5
user5@raspberrypi5:~$ hostname
raspberrypi5
user5@raspberrypi5:~$
```

```
user2@raspberrypi2: ~
File Edit Tabs Help
user2@raspberrypi2:~$ whoami
user2
user2@raspberrypi2:~$ hostname
raspberrypi2
user2@raspberrypi2:~$ mkdir ~/ShareRoot
bash: mkdir ~/ShareRoot: No such file or directory
user2@raspberrypi2:~$ ping -c 4 192.168.1.46
PING 192.168.1.46 (192.168.1.46) 56(84) bytes of data:
64 bytes from 192.168.1.46: icmp_seq=1 ttl=64 time=6.35 ms
64 bytes from 192.168.1.46: icmp_seq=2 ttl=64 time=6.19 ms
64 bytes from 192.168.1.46: icmp_seq=3 ttl=64 time=6.71 ms
64 bytes from 192.168.1.46: icmp_seq=4 ttl=64 time=32.7 ms

--- 192.168.1.46 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/mdev = 6.193/12.981/32.677/11.372 ms
user2@raspberrypi2:~$
```

SHARE

```
user1@raspberrypi1: ~/ShareRoot
File Edit Tabs Help
user1@raspberrypi1:~$ mkdir ~/ShareRoot
user1@raspberrypi1:~$ cd ~/ShareRoot/
user1@raspberrypi1:~/ShareRoot$ mkdir Folder1 Folder2 Folder3 Folder4 Folder5
user1@raspberrypi1:~/ShareRoot$ touch Folder1/doc1.txt Folder1/doc2.txt Folder2/
doc3.txt Folder3/doc4.txt Folder4/doc5.txt Folder5/doc6.txt Folder5/doc7.txt
user1@raspberrypi1:~/ShareRoot$ ls -R ~/ShareRoot
/home/user1/ShareRoot:
Folder1 Folder2 Folder3 Folder4 Folder5

/home/user1/ShareRoot/Folder1:
doc1.txt doc2.txt

/home/user1/ShareRoot/Folder2:
doc3.txt

/home/user1/ShareRoot/Folder3:
doc4.txt

/home/user1/ShareRoot/Folder4:
doc5.txt

/home/user1/ShareRoot/Folder5:
doc6.txt doc7.txt
user1@raspberrypi1:~/ShareRoot$
```

```
user1@raspberrypi1: ~/ShareRoot
File Edit Tabs Help
GNU nano 8.4 /etc/samba/smb.conf
comment = Printer Drivers
path = /var/lib/samba/printers
browseable = yes
read only = yes
guest ok = no
# Uncomment to allow remote administration of Windows print drivers.
# You may need to replace 'lpadmin' with the name of the group your
# admin users are members of.
# Please note that you also need to set appropriate Unix permissions
# to the drivers directory for these users to have write rights in it
; write list = root, @lpadmin

[ShareRoot]
path=/home/user1/ShareRoot
writeable=yes
browseable=yes
public=yes
create mask=0777
directory mask=0777

Help Write Out Where Is Cut Execute Location
Exit Read File Replace Paste Justify Go To Line
```

ON PC1, I CREATED A SHARED FOLDER CALLED
"SHAREROOT".
INSIDE THIS FOLDER, I CREATED 5 SUBFOLDERS AND
7 FILES.

THE FOLDER WAS CONFIGURED IN SAMBA AND
SHARED WITH READ AND WRITE PERMISSIONS TO
ALLOW ACCESS FROM OTHER NETWORK USERS.

Connectivity Test (Ping)

```
user2@raspberrypi2: ~  
File Edit Tabs Help  
user2@raspberrypi2:~$ whoami  
user2  
user2@raspberrypi2:~$ hostname  
raspberrypi2  
user2@raspberrypi2:~$ mkdir ~/ShareRoot  
bash: mkdir ~/ShareRoot: No such file or directory  
user2@raspberrypi2:~$ ping -c 4 192.168.1.46  
PING 192.168.1.46 (192.168.1.46) 56(84) bytes of data.  
64 bytes from 192.168.1.46: icmp_seq=1 ttl=64 time=6.35 ms  
64 bytes from 192.168.1.46: icmp_seq=2 ttl=64 time=6.19 ms  
64 bytes from 192.168.1.46: icmp_seq=3 ttl=64 time=6.71 ms  
64 bytes from 192.168.1.46: icmp_seq=4 ttl=64 time=32.7 ms  
  
--- 192.168.1.46 ping statistics ---  
4 packets transmitted, 4 received, 0% packet loss, time 3005ms  
rtt min/avg/max/mdev = 6.193/12.981/32.677/11.372 ms  
user2@raspberrypi2:~$
```

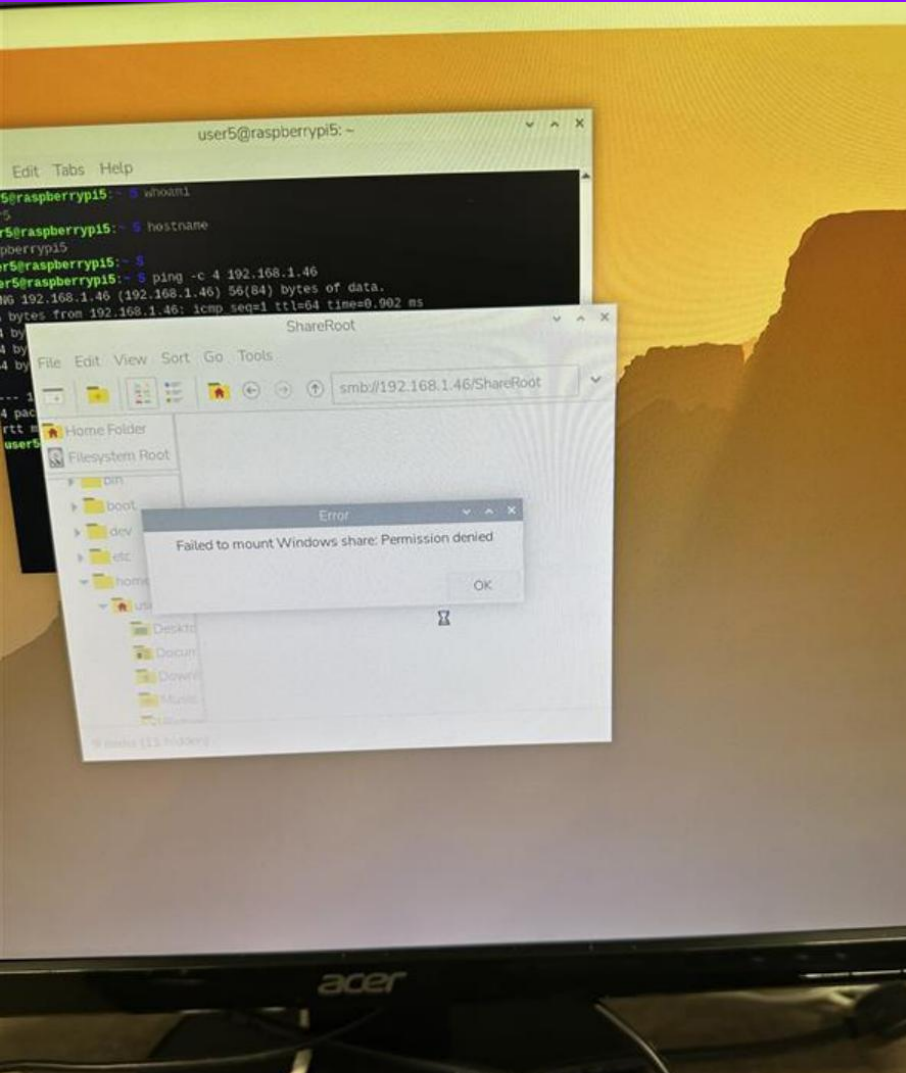
```
user5@raspberrypi5: ~  
File Edit Tabs Help  
user5@raspberrypi5:~$ whoami  
user5  
user5@raspberrypi5:~$ hostname  
raspberrypi5  
user5@raspberrypi5:~$ ping -c 4 192.168.1.46  
PING 192.168.1.46 (192.168.1.46) 56(84) bytes of data.  
64 bytes from 192.168.1.46: icmp_seq=1 ttl=64 time=0.902 ms  
64 bytes from 192.168.1.46: icmp_seq=2 ttl=64 time=0.879 ms  
64 bytes from 192.168.1.46: icmp_seq=3 ttl=64 time=0.875 ms  
64 bytes from 192.168.1.46: icmp_seq=4 ttl=64 time=0.862 ms  
  
--- 192.168.1.46 ping statistics ---  
4 packets transmitted, 4 received, 0% packet loss, time 3004ms  
rtt min/avg/max/mdev = 0.862/0.879/0.902/0.014 ms  
user5@raspberrypi5:~$
```

NETWORK CONNECTIVITY WAS
TESTED USING THE PING
COMMAND.

ALL DEVICES SUCCESSFULLY
COMMUNICATED WITH EACH
OTHER.

THE RESULTS SHOWED 0%
PACKET LOSS, CONFIRMING
THE NETWORK IS WORKING
CORRECTLY.

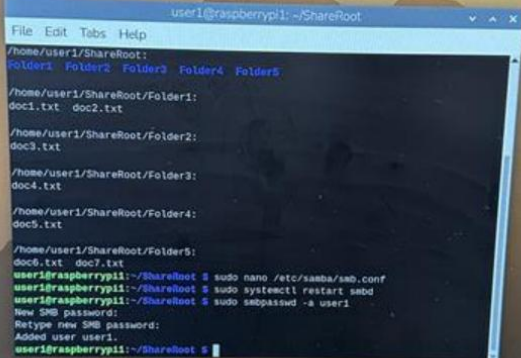
Initial Fault (Permission Denied)



WHEN ATTEMPTING TO ACCESS
THE SHARED FOLDER FROM
ANOTHER COMPUTER, ACCESS
WAS DENIED.
THIS WAS CAUSED BY MISSING
SAMBA AUTHENTICATION
PERMISSIONS.

FIX APPLIED (Samba Authentication Configured)

Samba password created and service restarted successfully.

A terminal window titled 'user1@raspberrypi1: ~/ShareRoot' showing a directory listing of folders and files, followed by commands to edit the Samba configuration file, restart the service, and create a new user with a password.

```
user1@raspberrypi1: ~/ShareRoot
File Edit Tabs Help
/home/user1/ShareRoot:
Folder1 Folder2 Folder3 Folder4 Folder5

/home/user1/ShareRoot/Folder1:
doc1.txt doc2.txt

/home/user1/ShareRoot/Folder2:
doc3.txt

/home/user1/ShareRoot/Folder3:
doc4.txt

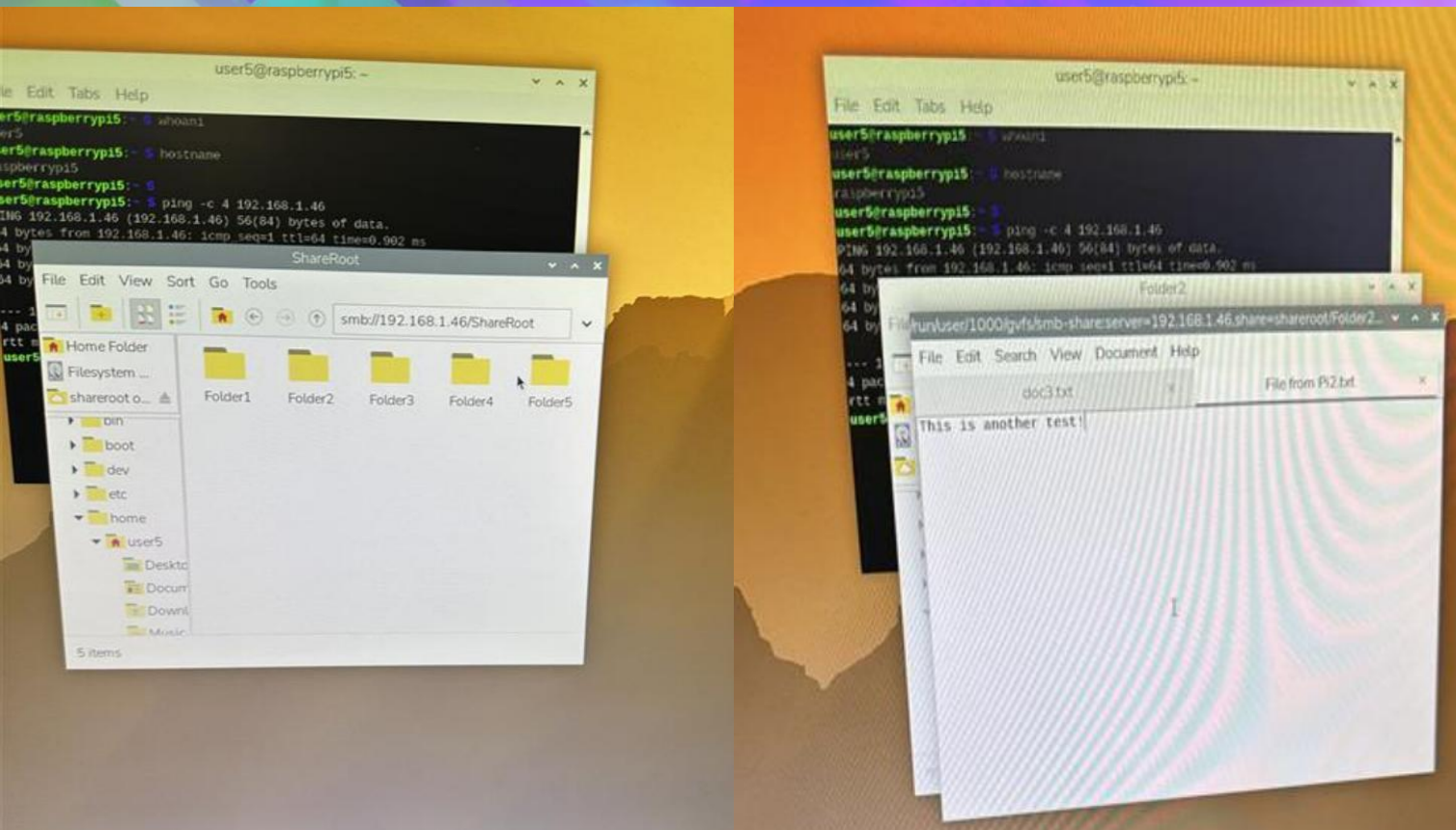
/home/user1/ShareRoot/Folder4:
doc5.txt

/home/user1/ShareRoot/Folder5:
doc6.txt doc7.txt
user1@raspberrypi1:~/ShareRoot $ sudo nano /etc/samba/smb.conf
user1@raspberrypi1:~/ShareRoot $ sudo systemctl restart samba
user1@raspberrypi1:~/ShareRoot $ sudo smbpasswd -a user1
New SMB password:
Retype new SMB password:
Added user user1.
user1@raspberrypi1:~/ShareRoot $
```

THE ISSUE WAS RESOLVED BY CREATING
A SAMBA PASSWORD FOR THE USER.
THE SAMBA SERVICE WAS THEN
RESTARTED TO APPLY THE CHANGES.
THIS ENABLED PROPER
AUTHENTICATION FOR NETWORK
ACCESS.

SUCCESSFUL ACCESS

Shared folder accessed successfully from another computer.



AFTER CONFIGURING
SAMBA
AUTHENTICATION, THE
SHARED FOLDER WAS
ACCESSED
SUCCESSFULLY FROM
ANOTHER DEVICE.
ALL SUBFOLDERS WERE
VISIBLE, CONFIRMING
CORRECT NETWORK
PERMISSIONS.

